Lang Cao

(+1) 217-621-0532 | windszzlang@gmail.com
Google Scholar | Homepage | GitHub

Education

University of Illinois at Urbana-Champaign (UIUC) Urbana, US
Master of Science in Computer Science (Research-based Program) *Aug. 2022 - May 2024*
- Research Area: AI for Healthcare; Natural Language Processing
- Advisor: Prof. Jimeng Sun
- Master Thesis: Automatic Rare Disease Extraction Based on Large Language Models
- GPA: 3.78/4.0

Wuhan University of Technology (WUT) Wuhan, China
Bachelor of Engineering in Software Engineering *Sept. 2018 - June 2022*
- Thesis Advisor: Prof. Jing Peng
- Bachelor Thesis: Automatic Argument Mining
- GPA: 4.351/5.0 (3.94/4.0); Rank: 1st/79

Publications

Learn to Refuse: Making Large Language Models More Controllable and Reliable through Knowledge Scope Limitation and Refusal Mechanism [Paper][Code]
- **Lang Cao** (Independent Research).
- **EMNLP 2024 Main Conference**, 2024 Conference on Empirical Methods in Natural Language Processing.

PILOT: Legal Case Outcome Prediction with Case Law [Paper] [Code]
- **Lang Cao**, Zifeng Wang, Cao Xiao, Jimeng Sun.
- **NAACL 2024 Main Conference**, 2024 Conference of the North American Chapter of the Association for Computational Linguistics.

KG-FIT: Knowledge Graph Fine-Tuning Upon Open-World Knowledge [Paper] [Code]
- Pengcheng Jiang, **Lang Cao**, Cao Xiao, Parminder Bhatia, Jimeng Sun, Jiawei Han
- **NeurIPS 2024**, The Thirty-Eighth Annual Conference on Neural Information Processing Systems.

GraphReason: Enhancing Reasoning Capabilities of Large Language Models through A Graph-Based Verification Approach [Paper][Code]
- **Lang Cao** (Independent Research).
- Natural Language Reasoning and Structured Explanations Workshop @ ACL 2024.

AutoRD: An Automatic and End-to-end Rare Disease Knowledge Graph Construction System Based on Ontologies-enhanced Large Language Models [Paper][Code]
- **Lang Cao**, Adam Cross, Jimeng Sun.
- JMIR Medical Informatics.

Accelerating Clinical Evidence Synthesis with Large Language Models [Paper]
- Zifeng Wang, **Lang Cao**, Benjamin Danek, Qiao Jin, Zhiyong Lu, Jimeng Sun.
- Under Review (npj Digital Medicine).

Adapted Large Language Models for Medical Literature Mining [Paper][Code]
- **Lang Cao**, Zifeng Wang, Qiao Jin, Zhiyong Lu, Jimeng Sun.
- Under Review (Nature Medicine).

AutoAM: An End-To-End Neural Model for Automatic and Universal Argument Mining [Paper][Code]
- **Lang Cao** (Independent Research).
- ADMA 2023, The 19th anniversary of the International Conference on Advanced Data Mining and Applications.

CBCP: A Method of Causality Extraction from Unstructured Financial Text [Paper][Code]
- **Lang Cao**, Shihua Zhang, Juxing Chen.
- NLPIR 2021, 2021 International Conference on Natural Language Processing and Information Retrieval.

DiagGPT: An LLM-based Chatbot with Automatic Topic Management for Task-Oriented Dialogue [Paper][Code]
- **Lang Cao** (Independent Research).
- Technical Report.

TabularThinking: Towards Language Models for Step-by-step and Human-Like Reasoning in Tables
- **Lang Cao**, Zhiyuan Lu, Haoyu Dong.
- In preparation.

Clustering of Functionally Related Genes Using Machine Learning Techniques [Paper]
- Yujing Xue, **Lang Cao**.
- ICCDA 2021, 2021 5th International Conference on Compute and Data Analysis.

Intelligent Cross-sensing Sensor Based on Deep Learning [Paper]
- Lingfei Xu, Jiaming Zhang, **Lang Cao**, Xinyu Hu.
- ICSIP2021, 2021 6th IEEE International Conference on Signal and Image Processing.

Experiences

Microsoft Beijing, China
Research Assistant at Microsoft AI & Microsoft Research (Data Knowledge and Intelligence Group) *Aug. 2024 – Now*
- Research Focus: Spreadsheet Intelligence and Table LLMs.
- Projects:
  - Dynamically Spreadsheet Encoding for Downstream Tasks
  - Spreadsheet Boundary Detection and Structure Extraction with Large Language Models
  - Language Model Reasoning with Tabular Data
- Mentor: Haoyu Dong

Tsinghua University, THUNLP Lab Beijing, China
Research Assistant *Nov. 2024 – Now*
- Research Focus: Multimodal Learning, Multimodal Large Language Model
- Projects:
  - Using Visual Chain-of thought to Enhance Multimodal Reasoning
  - Multimodal Alignment to Reduce Hallucination for Multimodal Large Language Models
- Mentor: Tianyu Yu

UIUC, Sunlab Urbana, US
Research Assistant *Jan. 2023 – Nov. 2024*
- Research Focus: Natural Language Processing for Applications in Healthcare and Legal
- Projects:
  - Legal Case Outcome Prediction with Case Law
  - Large Language Models for Rare Disease Information Extraction and Potential Rare Disease Prediction
  - Large Language Models for Automatic Clinical Trial
- Advisor: Jimeng Sun, Danica Xiao

AltairX (AI Startup at National University of Singapore) Kent Ridge, Singapore
Tech Leader / Tech Advisor / Researcher *May. 2024 – July. 2024*
- Explore emotional large language models, and AI for companion, human-computer interaction, and dialogue.
- Development of emotional companion applications and research on related core technologies.

Keiji (Healthcare AI Startup) Urbana, US
Research Intern *Jan. 2024 – May. 2024*
- Developed eligibility criteria generator demo, which can automatically generate patient eligibility criteria for clinical trials based on predefined parameters, thereby streamlining the trial setup process.
- Developed meta-analysis demo, which can automatically aggregate and analyze data from multiple studies, providing insights and trends that help in making informed decisions.

LegalNow (Legal AI Startup) Beijing, China
Tech Leader / Tech Advisor / Researcher *June 2023 – Sept. 2023*
- Individually completed the construction of an AI legal chatbot in the demo version of the product, which can assist and guide users in achieving multiple functions related to contracts drafting.

- Provided technical support and developed the overall AI framework for the first launching product.

iFLYTEK Hefei, China
NLP Algorithm Engineer at Smart Car Technology R&D Division *June 2021 - Aug. 2021*
- Smart Car Technology R&D Division
- Maintained and developed an automatic data iteration algorithm for training data of smart car AI system.
- Mentor: Shen'an Li

WUT, Bioinformatics Innovation Lab Wuhan, China
Research Assistant at Text Group *Sept. 2020 - Jan. 2021*
- Research Focus: Information Extraction and Text Mining.
- Projects: Named Entity Extraction from Biomedical Literature
- Advisor: Jing Peng

Software

PyHealth [Github] [PyHeath-GPT]
- A Deep Learning Python Toolkit for Healthcare Applications.
- Work Content:
  - Developed an AI chat assistant to help new users understand and learn how to use PyHealth.
  - Contributed code for the healthcare knowledge graph and the automation of clinical trial processes.

LegalNow AI [Homepage]
- A smart and friendly Lawyer-Grade AI for users to create or review legal documents.
- Work Content:
  - Designed and developed the overall AI framework to enable advanced task-oriented dialogues for legal applications.
  - Conducted prompt engineering and iteratively improved prompts to enhance system performance and usability.

Selected Honors & Awards

- Silver Medal, top 5% in Kaggle Common Lit Readability Prize (2021.8)
- Top 2% in Alibaba Tianchi NLP Chinese Pre-training Model Generalization Ability Challenge (2021.1)
- Silver Award in China College Students' "Internet Plus" Innovation and Entrepreneurship Contest (2020.9)
- 2nd Prize in National College Computer Ability Challenge - Artificial Intelligence Application Contest (2021.1)
- 3rd Prize in Service Outsourcing Innovation & Entrepreneurship Competition (SOIEC) for Chinese Students (2021.6)
- National Scholarship (1%), WUT (2020); Merit Student Model Honor (5‰), WUT (2020)
- Outstanding Graduate (2%), WUT (2022.6); Outstanding Thesis (1%), WUT (2022.6)
- **The National Champion** of the 2014 FIRST LEGO League Challenge in China (core) (2014.6); Gold Award at the Asia-Pacific Championship of the FIRST LEGO League Challenge (leader) (2%) (2016.7)

Skills

- Programming: Python, C/C++, Java, JavaScript/TypeScript, HTML/CSS, SQL, Shell, NXT-G/EV3-G
- Machine Learning Frameworks: PyTorch, TensorFlow, Keras, Huggingface Transformers, LangChain, DGL, Scikit-learn, NumPy, Pandas
- Software Development Frameworks: Vue.js, React.js, Node.js, Django, Flask, Spring Boot, Express, Hadoop, MFC, QT
- Tools: LaTeX, Markdown, Git, Linux, Docker

Services & Teaching

- Paper Review: ACL Rolling Review, ICLR, KDD, ICASSP
- Vice President, Artificial Intelligence Association, WUT (2020-2022)
- Teaching Assistant, Algorithm Design and Analysis, WUT (2020.2-2020.6)